

Proof. Observe that the flow line Γ ends at γ . Along Γ the braid representative goes from the one on the top left to the one on the top right of Figure 4.4, implying a crossing. This crossing is derived from $\tilde{q}$ and hence the claim for $(z_1, z_2) = (0, 0) \in [0, 1]_{z_1} \times [0, 1]_{z_2}$ holds. By continuity there exists a crossing derived from $\tilde{q}$ for each (z_1, z_2) in a small neighborhood of $(0, 0) \in [0, 1] \times [0, 1]$. $\square$

Let $U_{\tilde{q}}$ be the connected subset consisting of all $(z_1, z_2) \in [0, 1] \times [0, 1]$ satisfying Claim 4.6. For each $(z_1, z_2) \in U_{\tilde{q}}$ there exists an MFTS with 1 switching derived from $\tilde{q}$. Let $\Gamma(z_1, z_2)$ be the flow line starting at (z_1, z_2) and let $s = s_0$ be the moment given by Claim 4.6 when the crossing derived from $\tilde{q}$ occurs. We write $\Gamma(z_1, z_2)(s_0) = (\tilde{\gamma}'_1(z_1, z_2), \tilde{\gamma}'_2(z_1, z_2))$. Let $(\tilde{\gamma}''_1(z_1, z_2), \tilde{\gamma}''_2(z_1, z_2))$ be the result of applying the switching map; then $\tilde{\gamma}''_1(z_1, z_2)$ (resp. $\tilde{\gamma}''_2(z_1, z_2)$) is approximately the concatenation $a_1(z_1, z_2) \# b_2(z_1, z_2)$ (resp. $a_2(z_1, z_2) \# b_1(z_1, z_2)$), where we write $\tilde{\gamma}'_i(z_1, z_2) = a_i(z_1, z_2) \# b_i(z_1, z_2)$.

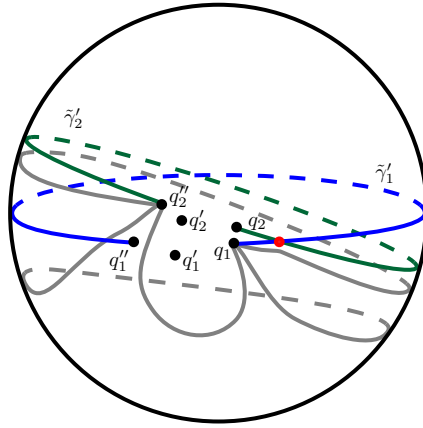


FIGURE 4.6. The curves $\tilde{\gamma}'_1$ (in blue) and $\tilde{\gamma}'_2$ (in green) intersect at the red dot, which represents the crossing derived from $\tilde{q}$. The curves in gray show various stages of the flow line starting at $\tilde{\gamma}'_1$.

We also claim that there exists a unique value z'_1 such that for $z_1 > z'_1$ there is no such MFTS with 1 switching starting at $(z_1, 0) \in [0, 1] \times [0, 1]$. Similarly there is a unique z'_2 with analogous property for the second coordinate. Moreover, on each ray starting from $(0, 0)$ there is a unique point satisfying analogous property and $U_{\tilde{q}}$ has the triangular shape as in Figure 4.5. The sides of this triangle along the z_1 - and z_2 -axes are denoted I_1 and I_2 and the hypotenuse is denoted I_3 .

Following the proof strategy of $\mu_M^2(x, 1) = x$ (see the paragraph on the verification of (1.6) above), we now analyze how the various MFTS approach the generator $(1, 1)$, represented by geodesics $g(q_1, q''_2)$ and $g(q_2, q''_1)$. For each of these generators there are two options: either a flow line approaches it from the right, or from the left. Hence there are 4 possible types of flow lines from (x_1, x_2) which we denote by rr, rl, lr, ll , where the first (resp. second) letter corresponds to the geodesic $g(q_1, q''_2)$ (resp. $g(q_2, q''_1)$).

Claim 4.7. *The pair $((1, \tilde{\gamma}_2), (\tilde{\gamma}_1, 1))$ corresponding to $(z_1, z_2) = (0, 0)$ is of type lr .*

Proof. Let $(\tilde{\gamma}'_1, \tilde{\gamma}'_2) = (\tilde{\gamma}'_1(0, 0), \tilde{\gamma}'_2(0, 0))$; see Figure 4.6. Note that $\tilde{\gamma}'_1$ and $\tilde{\gamma}'_2$ are very close to the generators x_1 and x_2 . Applying the switching map, $\tilde{\gamma}''_1 = \tilde{\gamma}''_1(0, 0)$ and $\tilde{\gamma}''_2 = \tilde{\gamma}''_2(0, 0)$ are approximately concatenations $a_1(0, 0) \# b_2(0, 0)$, $a_2(0, 0) \# b_1(0, 0)$ which are very close to two geodesics that intersect at an angle slightly less than π . The gradient vector at $\tilde{\gamma}'_1$, regarded as a vector field along $\tilde{\gamma}'_1$, has a relatively small norm away from this angle, whereas near the angle its norm is significantly greater than 0 and the vector points into the interior of this angle; see [Mil63, Chapter 12] for a more precise description. Hence $\tilde{\gamma}''_1$ is pushed away to the right along the flow line as shown in Figure 4.6 and the flow line